WOX7001 Research Methodology - Assignment 3

Temporal and Weather Feature Integration for Spatio-Temporal Graph Neural Networks Metro Ridership Prediction in Malaysia's Tropical Climate

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Research Background

Urbain rail transit
- Mobilise population

With accurate prediction:
- Resource allocation
- Service quality
- Efficient operations

Spatio-temporal Graph Neural Networks (STGNN)
- Spatial nature of metro network
- Temporal dynamic of hourly ridership

Integration of external features
- Temporal features
- Weather features

Problem Statement

Geographical concentration on temperate climate (USA and China), lack of empirical validation in tropical climate

Lack of ablation study to to identify the contribution of each features

Gap in feature importance analysis to understanding which temporal and weather features are the key drivers

Research Questions and Objectives

RQ1: What is the **baseline prediction** performance of STGNN using only historical data?
RO1: To evaluate performance of baseline STGNN architecture based on **standard regression metrics**.

RQ2: What is the effect to prediction performance **when temporal and weather features are integrated?**
RO2: To conduct **ablation study on the contribution** of temporal and weather features towards prediction performance

RQ3: Which specific temporal and weather features has the **strongest feature importance** in tropical climate ridership prediction?
RO3: To analyse the feature importance of temporal and weather features through **standard model interpretability methods.**

Literature Review

Architectural Success
- STGNNs effectively model metro ridership patterns[1]
- Combination of GCN (spatial) + GRU (temporal) = strong performance[2]

Feature Integration Trend:
- Multiple studies add temporal/weather features[1,3]
- Improvements reported in temperate climates[4]

Key Gaps:
- *Geographic*: Temperate climate dominance, zero tropical validation
- *Methodological*: No ablation studies isolating feature contributions
- *Analytical*: Unknown feature importance patterns in tropical contexts

Research Design

Data Collection (KTM)
- KTM Komuter ridership data from DOSM
- Temporal data
- Weather data

Data Preprocessing (pandas)
- Data cleaning
- Data transformation
- Feature engineering
- Graph structure engineering

Model Development (PyTorch)

Creation of model variants:
- Model Variant 1 (Ridership only)
- Model Variant 2 (M1+ Temporal)
- Model Variant 3 (M1 + M2 + Weather)

Split Train - Validation - Test set

Ablation and Feature Importance (Shap)
- Test set evaluation of baseline model M1 **(RO1)**
- Ablation analysis of M2 and M3 **(RO2)**
- Feature Importance of M2 and M3 **(RO3)**

Model Architecture

Input data — Komuter data, Temporal data, Weather data

GCN Layers — GCN Layer 1, GCN Layer 2 — Graph Convolutional Network layers to learn **spatial dependencies**

GRU Layer — Gated Recurrent Units to learn **temporal dependencies**

Fully Connected Layer — Final prediction output

Expected Outcome

RO1
Developed **benchmark metrics** for tropical climate.

RO2
Quantified impact of temporal and weather features contribution.

RO3
Identified **key predictors** in tropical settings.

11 SUSTAINABLE CITIES AND COMMUNITIES
- Improved transit prediction
- Better service quality

13 CLIMATE ACTION
- Optimised operations
- Reduced CO_2 emissions

References

[1] Jin, G., Liang, Y., Fang, Y., Shao, Z., Huang, J., Zhang, J., & Zheng, Y. (2023). Spatio-temporal graph neural networks for predictive learning in urban computing: A survey. IEEE transactions on knowledge and data engineering, 36(10), 5388-5408.
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[3] Zeb, M. S., Khan, M. A., Khattak, M. M. H., Ud-Din, S., Habib, M. F., & Khan, M. Z. (2025). Forecasting public transit ridership amidst COVID-19: a machine learning approach: MS Zeb et al. Public Transport, 17(2), 391-420.
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